DESIGN SYSTEMS

TECHNICAL DATA SHEET EPOPLAST A+B
SOLVENT-FREE COLOURED SELF-LEVELLING EPOXY FORMULATION

DESCRIPTION

ECOPLAST A+B is a two-component product based on epoxy resins and cycloaliphatic amine hardeners, for the construction of self-levelling coatings, skim coats and multi-layer systems. It can be applied by roller to produce anti-slip coatings and high-build, waterproof and spark-proof paint finishes.

Alternatively, ECOPLAST can be filled with quartz for skim-coat applications, increasing its impact and abrasion resistance. The formulation is supplied pre-coloured for some shades or in a non-pigmented (NP) version, with epoxy colour paste supplied separately.

USES

Waterproofing of channels and tanks. Floors for mechanical, food and chemical industries, depots and warehouses, laboratories and hospitals, shops, showrooms and apartments.

SUBSTRATE

The substrate must have a minimum compressive strength of 25 N/mm^2 and a tensile strength of 1,5 N/mm^2.

SUBSTRATE PREPARATION

Concrete substrates: they must be solid, dry (cured if newly laid), level, absorbent, not contaminated by oils, detergents, dust or other substances. Assess the most suitable type of mechanical preparation (grinding, shot-blasting or milling) and then apply a coat of PREPOX A+B. Any holes and minor defects can be repaired with fast-setting epoxy filler.

Tiled floors: they must be ground or shot-blasted until a completely matt surface is obtained; then apply one or more coats of PREPOX A+B or of a suitable fibre-reinforced primer (see the specific application cycle or contact the Design Systems technical service).

Existing resin coatings: they must be abraded or shot-blasted, then the dust residues removed.

Do not apply ECOPLAST on damp substrates or substrates subject to capillary rising damp (in this case contact the Design Systems technical service). The moisture content of the substrate must not exceed 4%.

APPLICATION

At the time of application, combine the two components in a single container and mix carefully for 2 minutes, using suitable equipment (drill with mixing paddle). Use the entire contents of the container quickly. When emptying the container, avoid scraping the sides and the bottom, as there may be product that is not perfectly blended.

Use in self-levelling cycles

The main application method of ECOPLAST is as a self-levelling coating. After mixing the two components, add QUARTZ B0 (0,8 kg per 1 kg of A+B) and mix again. Spread the product with a 5 mm notched trowel and,

within 5 minutes, go over it with a spiked roller using slow, regular movements to even out the surface. Consumption for a thickness of 2,5 mm is 2,2 kg/m^2 of A+B and 1,76 kg/m^2 of QUARTZ B0.

Use in skim-coat and multi-layer cycles

After mixing the two components, add QUARTZ B0 and mix again. The quantity of aggregate to be added is as follows:

- 0,8 kg per 1 kg of A+B for skim-coat cycles;
- 0,5 kg per 1 kg of A+B for multi-layer cycles.

Spread the product with fan-shaped movements, taking care not to leave excess material and evening it out with a roller while still wet. Consumption for each coat is:

- 0,35 kg/m^2 of A+B and 0,28 kg/m^2 of QUARTZ B0 for skim-coat cycles;
- 0,7 kg/m^2 of A+B and 0,35 kg/m^2 of QUARTZ B0 for multi-layer cycles.

Use in painting cycles

After mixing the two components, check the viscosity of the system, which varies greatly with temperature. To obtain anti-slip surfaces:

- apply a first coat of ECOPLAST (consumption 0,4 kg/m^2) and broadcast 1 kg/m^2 of QUARTZ B3 evenly;
- after 12-36 hours apply a second coat of ECOPLAST (consumption 0,5 kg/m^2).

For high-build paint finishes:

- apply ECOPLAST in two coats, for a total consumption of approx. 0,6 kg/m^2.

For painting, ECOPLAST can be diluted with universal solvent up to a maximum of 3% by weight (A+B).

WARNINGS

ECOPLAST coatings exposed to sunlight may fade or change colour, turning towards yellow: this does not in any way affect the performance of the coating. There may be slight differences between different production batches of the same colour: whenever possible, use material from the same batch.

For applications at low temperature, the material can be heated to 25°C to facilitate application and curing (reduction in viscosity).

Product for professional use: when applying the purchased product, the purchaser undertakes to strictly follow the warnings given above and the instructions in the safety data sheet.

PRODUCT DATA

DESCRIPTION	RESULT
Colour	As per price list or on request according to the RAL chart (minimum batch 200 kg)
Consumption as skim coat	0,35 kg/m^2 of A+B + 0,28 kg/m^2 of QUARTZ B0
Consumption as self-levelling coating	2,2 kg/m^2 of A+B + 1,76 kg/m^2 of QUARTZ B0 (thickness 2,5 mm)

Specific gravity (at 25°C)	A+B mix: 1,25 ± 0,05 g/ml; filled A+B mix: 1,60 ± 0,10 g/ml (with 80% QUARTZ B0)
Viscosity (at 25°C)	A+B mix: 800 ± 170 mPa·s (spindle 2, 30 rpm); filled A+B mix: 1.300 ± 700 mPa·s (with 80% QUARTZ B0, spindle 2, 6 rpm)
Solids content (A+B)	> 98%
VOC ready for use (Italian Legislative Decree 161/06)	< 200 g/l – Cat. A/j: two-component high-performance coating (BS)
Flash point	> 100°C
Tool cleaning	Universal solvent
Storage	12 months, in a dry place at a temperature between 5°C and 35°C

APPLICATION DATA AND TIMES

DESCRIPTION	RESULT
Mixing ratio (by weight)	Pre-coloured version: A = 100, B = 32; NP version: A = 95, paste = 5, B = 32
Pot life (50% RH)	at 15°C: > 40 min; at 25°C: 30 min
Touch dry (50% RH)	at 15°C: 12-16 hours; at 25°C: 5-7 hours
Foot traffic (50% RH)	at 25°C: 12 hours
Overcoating (50% RH)	at 25°C: from 12 to 36 hours
Full traffic (50% RH)	at 25°C: 36 hours

DESCRIPTION	RESULT
Full curing (50% RH)	at 25°C: 7 days
Ambient conditions for use	Temperature between +15°C and +35°C, RH < 50%, substrate moisture < 4% (*)
Maintenance of the coating	Use neutral detergents for cleaning

TECHNICAL PERFORMANCE DATA

DESCRIPTION	RESULT
Appearance	Glossy and flat
Gloss at 60°	95 (*)
Abrasion resistance (UNI 8298-9)	70-80 mg (Taber, CS-17 wheel, 1000 cycles, 1000 g)
Compressive strength (UNI 4279)	60 N/mm^2
Compressive modulus	1,5 GPa
Flexural strength (UNI 7219)	59 N/mm^2
Tensile strength (ASTM D 638)	40 N/mm^2

Hardness (ASTM D 2240)	Shore D 78
Adhesion (DIN ISO 4624)	> 1,5 N/mm^2
Coefficient of linear thermal expansion	20 × 10-6 °C-1
BCA wear resistance (EN 13892-4)	AR 0,5
Impact resistance (EN ISO 6272-1)	IR 10
Bond strength (EN 13892-8)	B2
Chemical resistance	Contact the Design Systems technical service

(*) ECOPLAST applied at substrate temperatures below 15°C may stain on contact with water or water-based preparations, forming whitish marks. This chemical resistance defect is caused by incomplete cross-linking. Therefore ECOPLAST must be applied with a substrate temperature of not less than 15°C and at least 3°C above the dew point.

The technical data given in this bulletin were obtained using standardised tests and standards, or modifications thereof, according to Design Systems systems. The methods applied may be requested from our technical service. The contents of this technical bulletin represent the best of our knowledge and experience. However, for uses other than those indicated, preliminary tests are recommended. In any case, Design Systems Srl accepts no responsibility for damage or defects caused by the use of its products, since the conditions of use are not under its control. It also advises that its technical service is available to customers for any information concerning the correct use of its products.